

VINAMRA VASHISHTH

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EDUCATION

Columbia University MS in Computer Science (Machine Learning Track) , 4.0/4.0 GPA LLM based Generative AI, Natural Language Processing, Cloud Computing, Analysis of Algorithms	New York, NY Dec 2026
Indian Institute of Information Technology Kota B.Tech. in Computer Science and Engineering , 3.95/4.0 GPA Neural Networks and Deep Learning, Machine Learning, Cloud Computing, Data Structures and Algorithms	Rajasthan, IN Aug 2024

WORK EXPERIENCE

MelioLabs Full Stack Data Engineer	Santa Clara, CA Jul 2024 - Aug 2025
- Accelerated the development of a scalable machine learning pipeline for efficient pre-processing and modeling of DNA melting curve data, facilitating accurate analysis and contributing to a 30% improvement in data processing efficiency - Engineered an intuitive web-based interface to seamlessly trigger post-processing ML pipelines on DNA melt data, with built-in live progress monitoring and log visualization, reducing data post-processing time by 25% and enabling a more accessible UX	
Full Stack Data Engineering Intern	Jan 2024 - Jun 2024
- Pioneered a data management system to automate workflows for 5,000+ high-throughput DNA melting curve datasets, cutting manual search time and study timelines by 50% with advanced search functionality and seamless access to analytical tools - Collaborated with cross-functional teams to transition tools infrastructure to cloud, leading to 40% reduction in tool downtime	
Software Development Intern	Dec 2022 - Dec 2023
- Architected and developed full-stack web applications for DNA melting curve analysis using React.js and Flask, processing and visualizing experimental data from 500+ DNA melting experiments with real-time interactive dashboards - Integrated clustering algorithms for representative curve analysis using K-means with dynamic time warping (DTW), automatically identifying distinct melt curve patterns per experiment with silhouette score validation	

PROJECTS

LLM-Powered Financial Strategy Translator and Backtester https://github.com/vv2418/llm-nl-backtester-genai	Sep 2025 - Dec 2025
- Developed an LLM-powered trading strategy translator which converts natural language rules into validated DSL and executes deterministic, hallucination-free backtests with reproducible execution logs and interpretable explanations - Built a tool-augmented financial backtesting pipeline leveraging LangGraph workflows to integrate market data, semantic parsing, schema validation, and performance metrics, ensuring 100% structural correctness across 89 prompt benchmarks	
C++ Data Structures and Algorithms Problem Solving Journal https://github.com/VinVash/450-DSA	Jul 2022 - Jul 2024
Designed and maintained a large-scale C++ repository focused on algorithmic problem solving, implementing performance-efficient solutions using modern C++ (STL, templates, smart pointers, OOP) for core data structures and algorithms	
Large-scale Time Series Annotation Platform https://github.com/VinVash/Major_Project_Time_Series	Mar 2023 - May 2023
Engineered a web application utilizing "Time series K-Means" clustering algorithm and "Uncertainty Sampling" active learning strategy to automate the annotation of a 10 million-point PCR melt curve dataset, cutting manual labeling time by 95%	

POSITIONS OF RESPONSIBILITY

Indian Institute of Information Technology Kota Lead Organizer of "Hack The Chain" hackathon	Rajasthan, IN Mar 2023 - Mar 2023
Managed a global hackathon with 800+ registrations, outlining success metrics, securing 15+ sponsors and INR 150K prize pool	
Zine Robotics and Research Club Core Team Member	Rajasthan, IN Feb 2021 - Mar 2022
Mentored 100+ students in a robotics workshop on topics such as PID control, Aerodynamics and Arduino programming	

SKILLS

Machine Learning and Deep Learning: Python, Pytorch, Tensorflow, Scikit-learn, Numpy, Pandas
LLM & Generative AI: LangChain, LangGraph (RAG pipelines), LoRA (Fine-tuning), OpenAI Python library, HuggingFace library
Web Development: JavaScript, TypeScript, React.js, Redux Toolkit, Next.js, Node.js, Express.js, Flask, FastAPI, MongoDB, Mongoose
Systems: C/C++, UNIX shell scripting, Git, Docker, Blockchain